

1) solve the following problem with CPU scheduling Algorithms

Draw a table with 4 rows with

| Process        | AT | BT |
|----------------|----|----|
| P <sub>1</sub> | 0  | 5  |
| P <sub>2</sub> | 0  | 4  |
| P <sub>3</sub> | 0  | 3  |

If there is a context switch in the CPU scheduling Algorithm then there is a context switch time of 5 sec so each unit is considered as seconds.

Two Scheduling Algorithms

- 1) Preemptive      2) Non-Preemptive.

Once CPU is assigned with anyone of the

opp of preemptive

Process

Same.

3 Scheduling Algorithms

1) SRTF

shortest Run...

Time first

1) FCFS (First come first serve)

2) SJF (Shortest Job first)

3) PS (Priority scheduling)

2) PS

3) RR (round robbing)

for given problems

If 4 columns have we can apply all 6 algorithms

(Process, AT, BT, Priority)

4 columns

If last column is not there we can only use 4

algorithms (PS x)

(Process, AT, BT)



if RT is not given, we have to assume as 1

FCFS:

- Arrival of process

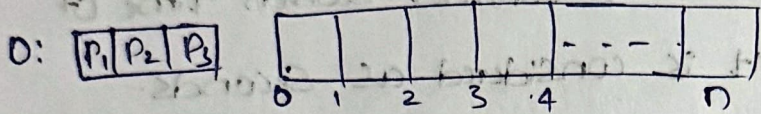
- All three process are arrived at same time that is 0.

RA → Data structure shows process which are available for execution.

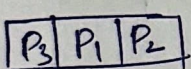
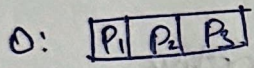
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RA

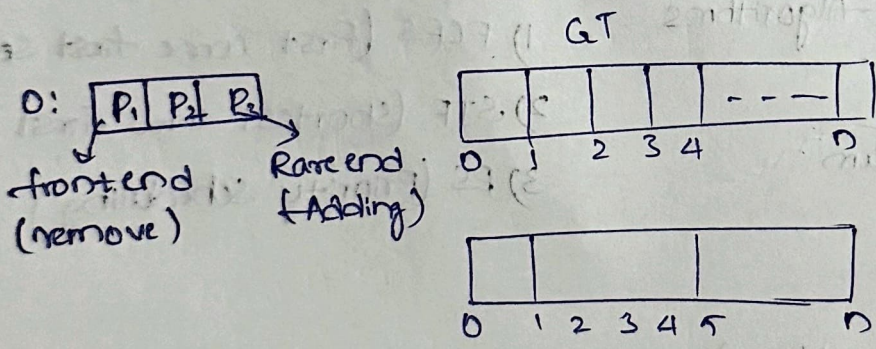
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If all arrive at same then consider least bus time also

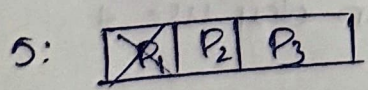


because all three arrived at same time

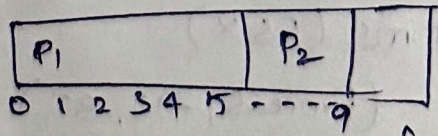
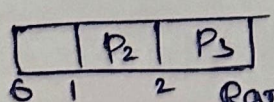


If we apply CPU to process it is remained with process until the process execution is completed (Non-preemptive)

- Amount of Time required for completion of process execution is called Bus Time



GT



frontend = 1      Rear index = 2

4 units required for P<sub>2</sub>

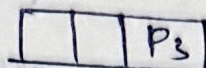
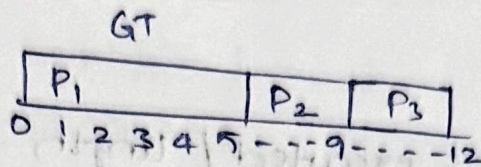




$F = 1 + 1 = 2, R = 2$

$F = 2 = R = 2$  so

the list/queue is having one element

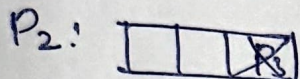


$F = R = \text{equals}$

$-1 \Rightarrow$  queue empty

$P_3$  is completed at 12 second

$P_3$  is removed there



$F = 2, R = 2$

now  $F = R = 1$

queue is empty

when queue is empty 2 situations might occur:

1) All process are completed

2) During that time CPU

Using GT calculate CP completion time, TAT, WT & RT Response time.

| Turn Around Time | Waiting time CT | TAT CT+AT | WT TAT-BT | RT |
|------------------|-----------------|-----------|-----------|----|
| $P_1$            | 5               | 5         | 0         | 0  |
| $P_2$            | 9               | 9         | 5         | 5  |
| $P_3$            | 12              | 12        | 9         | 9  |

When CPU is assigned for

RT

for Non-preemptive, WT & RT are same

CPU length:

Ending position of CPU - Starting position of CPU

$12 - 0 = 12$



## Throughput:

$$\text{No. of process executed / length} = 3/2 = 0.25$$

per unit 0.25 no. of process are executed.

maximizing Throughput & minimizing WT & RT

$$\text{Avg} = \frac{26}{3} \quad \frac{\text{Sum of TAT}}{\text{No. of process}}$$

$$= \frac{\text{Sum of WT}}{\text{No. of process}} = \frac{14}{3}$$

⇒ Is convection & starvation are applied to it or not.

⇒ Is context switeling is happening or not.

Which gives highest throughput & lowest Avg TAT & WT we can choose Suitable Algorithm.

## SRTF:

Shortest Remaining Time First

Context Switch:

RR

(shortest BT)

0: 

|                |                |                |
|----------------|----------------|----------------|
| P <sub>3</sub> | P <sub>2</sub> | P <sub>1</sub> |
|----------------|----------------|----------------|

|                |  |
|----------------|--|
| P <sub>3</sub> |  |
|----------------|--|

0 1

executed by 1 unit out of 3 remaining = 2

1: 

|                |                |                |
|----------------|----------------|----------------|
| P <sub>3</sub> | P <sub>2</sub> | P <sub>1</sub> |
|----------------|----------------|----------------|

acc for ReT update RR

$$\text{ReT}(P_3) = \text{BT} - \text{current execution Time}(P_3) \\ 3 - 1 = 2$$

1: 

|                |                |                |
|----------------|----------------|----------------|
| P <sub>3</sub> | P <sub>2</sub> | P <sub>1</sub> |
|----------------|----------------|----------------|

$$\text{ReT}(P_2) = 4 - 0 = 4$$

$$\text{no updation} \quad \text{ReT}(P_1) = 5 - 0 = 5$$

2: 

|                |                |                |
|----------------|----------------|----------------|
| P <sub>3</sub> | P <sub>2</sub> | P <sub>1</sub> |
| 1              | 4              | 5              |

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|                |                |   |
|----------------|----------------|---|
| P <sub>3</sub> | P <sub>3</sub> |   |
| 0              | 1              | 2 |



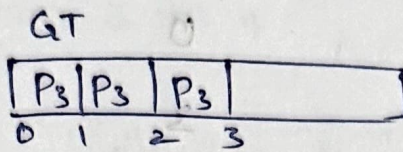
$$RT(P_3) = 1$$

$$RT(P_2) = 4$$

$$RT(P_1) = 5$$

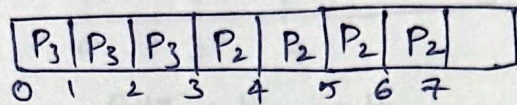
again select shortest RT that is  $P_3$

3:  $P_3 | P_2 | P_1$



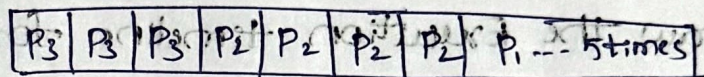
$RT(P_3) = 0 \Rightarrow$  Process execution is completed

4:



at  $t=7$   $P_2$  completed

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PROCESS CT TAT WT RT

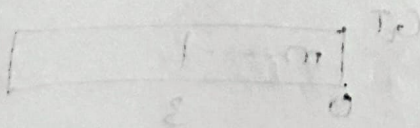
$P_1$  12 12 7

$P_2$  7 7 3

$P_3$  3 3 0

FCFS & SRTF

SRTF is best because giving less WT



$$WT = 2 + 0 + 0 = 2$$



10) Real-life CPU scheduling scenario:

→ A mobile operating system has following tasks

| Task                  | AT | CPU Burst |
|-----------------------|----|-----------|
| Music Player          | 0  | 20        |
| Touch Event           | 3  | 1         |
| Whatsapp Notification | 5  | 2         |
| Camera Launch         | 6  | 4         |

Questions:

- 1) Which scheduling algorithm is most suitable?  
FCFS, SJF, SRTF or RR?
- 2) Draw the Execution timeline for SRTF
- 3) Explain why the selected algorithm improve user experience.
- 4) Which task receives the fastest response time?

Ans

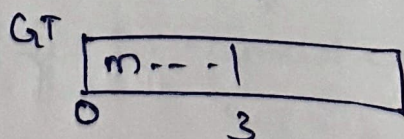
1) Suitable algorithm is SRTF because always checks the if a newly arrived task has a shorter remaining execution time than currently running task.

2) Execution timeline for SRTF is

at Time 0:

m = music player

AT = 3



$$Ret(mp) = 20 - 3 = 17$$

Bus time - Current Execution Time



at Time 3:

T = Touch event

arrives at 3.

BT = 1

TE runs from 3 to 4

$$Re(T) = 17$$

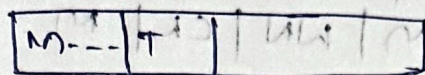
BT of TE(1)

$$TE \leq Re(M)$$

at Time 4:

Touch event finishes

GT



$$Re(T) = 17 - 1 = 16$$

at Time 5:

Whatsapp Notification

arrival Time = 5

BT = 2

$$2 < 16$$

Next AT = 6

at 6 WN has run for unit

$Re(WN) = 1$  Camera launch arrives

BT = 4

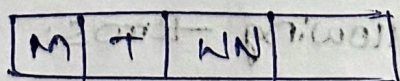
Now compare

$Re(MP)(16)$  VS  $Re(WN)(1)$  VS  $CL(A)$

Continue with WN

WN runs from 5 to 7

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at time 7 (WN) finishes

Now compare  $Re(M)$  16 VS  $CL(A)$

CL is shorter so allocating CPU to CL



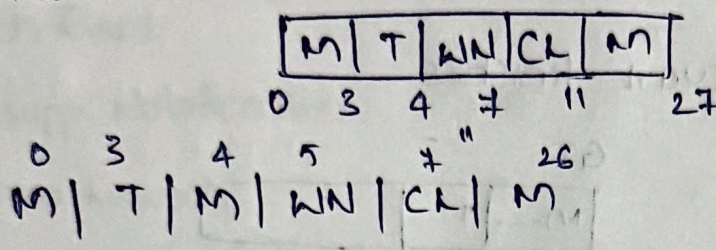
CL runs from 7 to 11

at time 11 CL finishes

now allocating CPU to MP

MP runs continuously until completion

$$11 + 16 = 27 \quad \text{GT}$$



MP  $\rightarrow$  TE  $\rightarrow$  WN  $\rightarrow$  CL  $\rightarrow$  MP

3) SRTF give a quick response to small task like touch events and notifications. so, the mobile feels fast and responsive.

4) Touch

| Task | AT | BT | first CPU | CT | TAT<br>CT-AT | WT<br>TAT-BT | RT |
|------|----|----|-----------|----|--------------|--------------|----|
| MP   | 0  | 20 | 0         | 27 | 27           | 7            | 0  |
| TE   | 3  | 1  | 3         | 4  | 1            | 0            | 0  |
| WN   | 5  | 2  | 5         | 7  | 2            | 0            | 0  |
| CL   | 6  | 4  | 7         | 11 | 5            | 0            | 0  |

Music player, Touch Event & Whatsapp Notification  
Share the fastest response time of 0.

3) Define the following terms:

Starvation, Convey effect, Context switching,  
Aging.



1) Starvation: It occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both preemptive & non-preemptive scheduling, depending on the algorithm.

Ex: A low priority process may have to keep waiting while high priority process are continuously executed.

2) Convoy Effect: It is an important problem associated with non-preemptive scheduling, especially FCFS. The long process acts like a convoy, causing many short processes to follow behind it.

3) Context Switching

The PCB stores the current execution state of a process, including the program counter, CPU registers, stack pointer & other CPU related information. When the CPU switches from one process to another, the OS saves the state of currently running process in its PCB & loads the saved state of the next process from its PCB. This allows the interrupted process to resume execution from exactly where it was paused.

- Aging: It is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.



| Process        | AT | BT | Priority |
|----------------|----|----|----------|
| P <sub>1</sub> | 0  | 4  | 3        |
| P <sub>2</sub> | 0  | 9  | 1        |
| P <sub>3</sub> | 0  | 6  | 2        |
| P <sub>4</sub> | 0  | 4  | 3        |
| P <sub>5</sub> | 0  | 9  | 1        |
| P <sub>6</sub> | 0  | 6  | 2        |
| P <sub>7</sub> | 1  | 4  | 3        |
| P <sub>8</sub> | 1  | 9  | 1        |
| P <sub>9</sub> | 2  | 6  | 2        |

Apply all the CPU scheduling algorithms for insert.

FCFS : First come first serve

Req 0:

P<sub>1</sub> P<sub>2</sub> P<sub>3</sub> P<sub>4</sub> P<sub>5</sub> P<sub>6</sub>

1: P<sub>1</sub> P<sub>2</sub> P<sub>3</sub> P<sub>4</sub> P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub>

2: P<sub>1</sub> P<sub>2</sub> P<sub>3</sub> P<sub>4</sub> P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>

4: ~~P<sub>1</sub>~~ P<sub>2</sub> P<sub>3</sub> P<sub>4</sub> P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>

13: ~~P<sub>2</sub>~~ P<sub>3</sub> P<sub>4</sub> P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>

19: ~~P<sub>3</sub>~~ P<sub>4</sub> P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>

23: ~~P<sub>4</sub>~~ P<sub>5</sub> P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>

32: ~~P<sub>5</sub>~~ P<sub>6</sub> P<sub>7</sub> P<sub>8</sub> P<sub>9</sub>



38: 

|               |    |    |    |
|---------------|----|----|----|
| <del>P6</del> | P7 | P8 | P9 |
|---------------|----|----|----|

42: 

|               |    |    |
|---------------|----|----|
| <del>P7</del> | P8 | P9 |
|---------------|----|----|

51: 

|               |  |
|---------------|--|
| <del>P8</del> |  |
|---------------|--|

57: 

|               |
|---------------|
| <del>P9</del> |
|---------------|

GT:

|       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ | $P_7$ | $P_8$ | $P_9$ |
| 0     | 4     | 13    | 19    | 23    | 32    | 38    | 42    | 51    |

| Process        | AT | BT | CT | TAT<br>CT-AT | WT<br>TAT-BT | RT<br>WT-AT |
|----------------|----|----|----|--------------|--------------|-------------|
| P <sub>1</sub> | 0  | 4  | 4  | 4            | 0            | 0           |
| P <sub>2</sub> | 0  | 9  | 13 | 13           | 4            | 4           |
| P <sub>3</sub> | 0  | 6  | 19 | 19           | 7            | 7           |
| P <sub>4</sub> | 0  | 4  | 23 | 23           | 19           | 19          |
| P <sub>5</sub> | 0  | 9  | 32 | 32           | 23           | 23          |
| P <sub>6</sub> | 0  | 6  | 38 | 38           | 32           | 32          |
| P <sub>7</sub> | 1  | 4  | 42 | 42           | 37           | 36          |
| P <sub>8</sub> | 1  | 9  | 51 | 50           | 41           | 40          |
| P <sub>9</sub> | 2  | 6  | 57 | 55           | 49           | 47          |